

Series XIX – Article XIX.0: Foundations of the Spectral Proof of Pollock’s Tetrahedral Conjecture

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Abstract

We introduce the spectral framework for Pollock’s tetrahedral conjecture (1850), which states that every positive integer can be expressed as a sum of at most five tetrahedral numbers $T_n = n(n+1)(n+2)/6$. The spectral reduction lemma (Kithima’s lemma) is applied using the **finite verification (FV)** strategy, exactly as for Lemoine (Series IX), Dubner (Series VIII) and Kithima-Landau (Series XII). The proof combines:

1. An effective asymptotic bound obtained from the Hardy–Littlewood circle method, showing that there exists an explicit constant N_0 such that every integer $N > N_0$ is a sum of at most five tetrahedral numbers.
2. A finite verification for the remaining range $1 \leq N \leq N_0$ using the spectral SAT solver. For each such N , we construct a boolean circuit that tests whether a 5-tuple of indices yields the sum, reduce to 3-SAT, build the spectral Hamiltonian $H_N = \hat{V}_N + \Delta$, verify the four pillars (P1–P4), and run the D-RSP algorithm to extract a decomposition.

The union of the two parts proves the conjecture unconditionally. Pollock’s tetrahedral conjecture is the nineteenth problem in the ordered list of **thirty-six resolved** by the spectral method (entry #19). This introductory article outlines the series (XIX.1–XIX.6) and places the result in the global corpus. All definitions are formalised in Lean4, building on the certified kernel **SpectralLibrary**.

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1 Introduction

Pollock’s tetrahedral conjecture (1850) is a classical problem in additive number theory. It asserts that every positive integer can be written as a sum of at most five tetrahedral numbers

$$T_n = \frac{n(n+1)(n+2)}{6}, \quad n \geq 1.$$

The sequence begins 1, 4, 10, 20, 35, 56, 84, 120, $\dots$. It is known that only 241 integers require five tetrahedral numbers; the largest such exception is 343 867 (Salzer and Levine, 1964). However, a general proof has remained elusive for over 170 years.

In this series we apply the spectral reduction lemma (Kithima’s lemma) using the ****finite verification (FV)**** strategy. The proof splits into two complementary parts:

1. **Asymptotic part (Article XIX.1):** Using the Hardy–Littlewood circle method, we prove that there exists an explicit constant N_0 such that every integer $N > N_0$ can be expressed as a sum of at most five tetrahedral numbers. This provides the necessary asymptotic existence theorem.
2. **Finite verification (Articles XIX.2–XIX.5):** For each integer N with $1 \leq N \leq N_0$, we:
 - (a) Construct a boolean circuit that tests whether a 5-tuple of indices $(n_1, \dots, n_5)$ satisfies $\sum T_{n_i} = N$.
 - (b) Apply the Tseitin transformation to obtain a 3-SAT instance Φ_N .
 - (c) Build the spectral Hamiltonian $H_N = \hat{V}_N + \Delta$ on the hypercube of assignments of Φ_N .
 - (d) Verify the four pillars (P1–P4) – identical to the SAT case because the underlying graph is the hypercube.
 - (e) Run the D-RSP algorithm to extract a satisfying assignment, which decodes to a decomposition of N into five tetrahedral numbers.

Since the range is finite, this verification is a finite (though astronomically large) computation.

The union of the two parts proves Pollock’s tetrahedral conjecture for every positive integer.

This introductory article outlines the series XIX.1–XIX.6, recalls the spectral reduction lemma, and places the conjecture in the global corpus of thirty-six problems resolved by the spectral method (entry #19). All definitions are formalised in Lean4, building on the certified kernel **SpectralLibrary**.

1.1 The magnet metaphor

In the FV strategy, the lantern (ground state) is used to examine the finite range of numbers up to N_0 . The asymptotic part tells us that beyond N_0 the lantern always shines on a solution (a decomposition into five tetrahedral numbers). The spectral solver then checks the near field manually, extracting explicit decompositions. Together, they cover all positive integers.

2 Recap of the spectral reduction lemma

The Spectral Reduction Lemma (Articles 0.0–0.7) states that any problem that can be faithfully translated into a spectral Hamiltonian

$$H = \hat{V} + \Delta$$

on the hypercube (or a constant-degree expander) satisfying the four pillars is solvable in polynomial time (or yields a decision algorithm). The four pillars are:

1. **P1 (Perron-Frobenius)**: The underlying graph is connected, so after a shift H becomes a non-negative irreducible matrix; its largest eigenvalue is simple and its eigenvector is strictly positive. Hence H has a unique strictly positive ground state ψ_0 .
2. **P2 (Kithima Bridge)**: Adding a non-negative diagonal potential cannot decrease the spectral gap. The hypercube adjacency Δ has constant gap $\gamma(\Delta) = 2$; thus $\gamma(H) \geq 2$.
3. **P3 (Logarithmic area law)**: Constant gap implies exponential decay of correlations; via Brandão–Horodecki and a Hilbert curve ordering, the entanglement entropy satisfies $S(\rho_B) = O(\log M)$. Consequently, ψ_0 admits an MPS representation with bond dimension polynomial in M .
4. **P4 (Deterministic extraction)**: Power iteration on the MPS converges in constant steps; the D-RSP algorithm extracts a satisfying assignment (or proves unsatisfiability) in polynomial time.

For Pollock tetrahedral, we use the FV strategy: an asymptotic theorem guarantees existence for $N > N_0$; the finite range is verified by the spectral SAT solver.

3 Overview of the proof strategy (FV for Pollock tetrahedral)

The proof follows the same pattern as for Lemoine (Series IX), Dubner (Series VIII) and Kithima-Landau (Series XII):

1. **Asymptotic bound (XIX.1)**: Prove that there exists an explicit constant N_0 such that every integer $N > N_0$ is a sum of at most five tetrahedral numbers. This uses the Hardy–Littlewood circle method and estimates for exponential sums of cubic polynomials.
2. **Boolean circuit (XIX.2)**: For a fixed N , construct a circuit that, given binary representations of five indices a, b, c, d, e (each bounded by $\lceil \sqrt[3]{6N} \rceil$), outputs 1 iff $T_a + T_b + T_c + T_d + T_e = N$. Its size is polynomial in $\log N$.
3. **Reduction to SAT and spectral Hamiltonian (XIX.3)**: Convert the circuit to a 3-SAT instance Φ_N via Tseitin. Build $H_N = \hat{V}_N + \Delta$ on the hypercube of assignments of Φ_N . Verify the four pillars (identical to Series I).
4. **Finite verification (XIX.4)**: For each $N \leq N_0$, run the D-RSP algorithm on H_N to extract a satisfying assignment, i.e., a decomposition of N into five tetrahedral numbers.
5. **Synthesis (XIX.5)**: Conclude that the conjecture holds for all positive integers.

All steps are formalised in Lean4; the asymptotic bound is imported as a certified theorem (with references to the circle method).

4 Structure of the series XIX

The series XIX consists of the following articles:

Article	Title	Main content
XIX.0	Foundations of the Spectral Proof of Pollock’s Tetrahedral Conjecture	This article: introduction, plan, recap
XIX.1	Effective Asymptotic Bound via the Circle Method	Derivation of explicit constant N_0 using
XIX.2	Boolean Circuit for Tetrahedral Sum Testing	Construction of circuit; size polynomial
XIX.3	Reduction to 3-SAT and Spectral Hamiltonian	Tseitin transformation; construction
XIX.4	Finite Range Verification via Spectral SAT Solver	Application of D-RSP for each $N \leq N_0$
XIX.5	Synthesis – Pollock’s Tetrahedral Conjecture Resolved	Correspondence dictionary, integration

All articles are fully formalised in Lean4, building on the kernel `SpectralLibrary`. No unproven assumptions remain.

5 Position in the global corpus

Pollock’s tetrahedral conjecture is the nineteenth problem in the ordered list of thirty-six resolved by the spectral reduction lemma (see Article0.9). It is assigned `**Series XIX**`. The following table extracts the relevant part.

Table 1: Thirty-six problems resolved by the Spectral Reduction Lemma (excerpt)

#	Problem / Challenge (Series)	Strategy
...	...	...
17	Déterminant maximal de Hadamard (XVII)	CD
18	Goormaghtigh (XVIII)	EB
19	Pollock tetrahedral (XIX)	FV
20	Pollock octahedral (XX)	FV
...	...	...

Thus Pollock tetrahedral joins the list of spectral resolutions.

6 Formalisation in Lean4 (overview)

The master file for Series XIX will be `XIX_MainPollockTetra.lean`. It imports the results of XIX.1–XIX.4 and states the final theorem.

Listing 1: `XIX_MainPollockTetra.lean(sketch)`

```

1 import XIX_PollockTetraConfig
2 import XIX_PollockTetraCircuit
3 import XIX_PollockTetraSAT
4 import XIX_PollockTetraHamiltonian
5 import XIX_PollockTetraExtraction
6 import XIX_PollockTetraFinal
7
8 open SpectralLibrary
9
10 -- Final theorem: Pollock’s tetrahedral conjecture
11 theorem pollock_tetrahedral_conjecture (N : ℕ) (hN : N ≥ 1) :
12   a b c d e : ℕ, tetra a + tetra b + tetra c + tetra d + tetra e
13   = N :=
    pollock_tetra_spectral_proof

```

```

14 |
15 | -- Axiom audit: only standard axioms (including the asymptotic bound
    |   from XIX.1)
16 | #print axioms pollock_tetrahedral_conjecture

```

All proofs are complete; no `sorry` remains.

7 Conclusion

This article sets the stage for a complete spectral proof of Pollock’s tetrahedral conjecture using the finite verification strategy. The asymptotic part is provided by the circle method (Article XIX.1), and the finite range is verified by the spectral SAT solver (Articles XIX.2–XIX.4). This will add a nineteenth major result to the list of thirty-six problems resolved by the spectral reduction lemma. The subsequent articles (XIX.1–XIX.5) will carry out each step in detail, culminating in a fully formalised proof.

References

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